SHIELDPATCH

PREDICT · PROTECT · PREVAIL

AI-Powered Vulnerability Detection & Automated Patch Management System

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01 ABSTRACT & OVERVIEW

ShieldPatch is an intelligent cybersecurity platform that automates vulnerability detection and patch management using ML-based risk prediction, real-time threat intelligence, and Docker sandbox testing.

⚠ **PROBLEM**

Manual patching is slow, error-prone & leaves systems exposed to known CVEs and malware attacks.

✓ **SOLUTION**

Automated ML-driven pipeline that detects, scores, recommends & applies patches safely.

02 METHODOLOGY

1. **Scan & Detect** — System vulnerability detection
2. **Threat Intelligence** — NVD, EPSS, ExploitDB feeds
3. **ML Risk Engine** — Risk prediction (R^2=0.81)
4. **Prioritize** — Rank patches by risk score
5. **Test & Deploy** — Sandbox testing + rollback
6. **Monitor** — Real-time dashboard alerts

03 TECHNOLOGIES USED

Python 3.9	React 18.2	Flask 2.3	MySQL 8.0
Sklearn 1.3	Docker 25.0	NVD / EPSS	Rasa 3.5

04 SYSTEM ARCHITECTURE

PRESENTATION — React.js UI, Dashboard, AI Chatbot ⇄ **BUSINESS** — Auth & Access, File Upload, Routing ⇄ **SERVICE LAYER** — Scan Engine, Threat Intel, ML Prediction, Patch Module ⇄ **DATA** — MySQL DB, Docker Sandbox

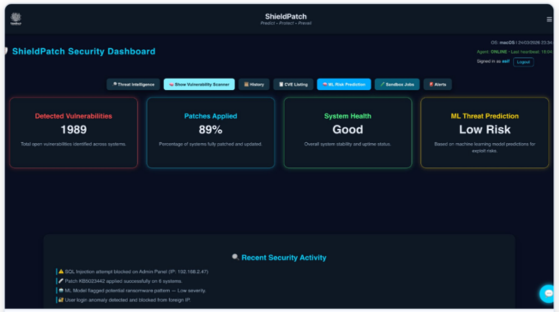
05 RESULTS & SCREENSHOTS

1989 VULNERABILITIES DETECTED

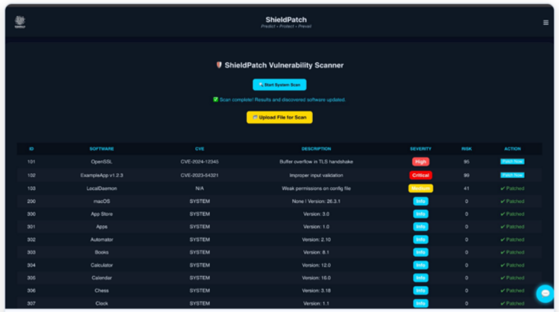
89% PATCHES SUCCESSFULLY APPLIED

0.81 ML MODEL R^2 ACCURACY

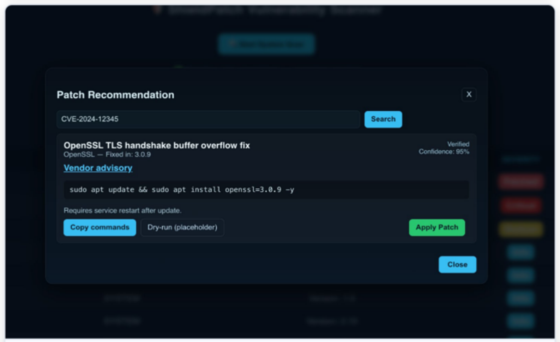
LOW RISK ML THREAT PREDICTION



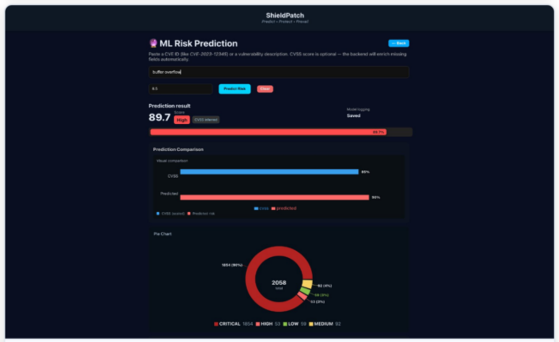
🖥 Security Dashboard



🔍 Vulnerability Scanner



🩹 Patch Recommendation



🤖 ML Risk Prediction

06 CONCLUSION

ShieldPatch successfully bridges the gap in automated vulnerability management by combining real-time threat intelligence, ML-based risk scoring, and Docker sandbox patching into one unified platform — reducing manual effort and improving security response time.

🚀 **FUTURE ENHANCEMENTS**

- Larger ML Dataset
- Auto-Patch Scheduling
- Cloud Deployment
- Multi-Agent Orchestration